EFY TESTED S.C. Dwivedi

Condensed Water Atomiser For Air-conditioners

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As the world gets hot due to global warming, air-conditioners are becoming a necessity to keep cool at home and workplaces. But at times it becomes difficult to provide proper drainage for the water condensed from air by the air-conditioners. It is not unusual, in absence of proper drainage, to collect the condensed water in buckets and throw it from time to time. This project aims to solve the problem.

The presented system can be installed in an air-conditioner's water outlet, which should remain outside the air-conditioned room. The ultrasonic atomiser used here converts the condensed water collecting in the air-conditioner's small tray into small aerosol particles using the mechanical energy produced by ultrasonic waves.

Circuit and working

Circuit diagram of the condensed water atomiser is shown in Fig. 1. It is built around 15V step-down transformer X1, bridge rectifier BR1, 12V voltage regulator 7812 (IC1), Arduino Uno Board1, two 5mm LEDs—LED1 and LED2, 2N2222 transistor T1, 12V relay RL1, ultrasonic atomiser disk L1, and water-level sensor SEN18.

Arduino Uno. Arduino Uno board is connected to water-level sensor SEN18, which checks water level and gives signal to Arduino. The Arduino program compares the signal with the set value in program. If the water level is above the set value, relay RL1 is energised through Arduino pin 13 and the ultrasonic atomiser is turned on. The program turns off the ultrasonic atomiser by de-energising the relay when water level reaches below the set point.

Power supply. It provides power to the circuit and to the relay driver used for switching on the ultrasonic atomiser. It comprises 15V, 2A step-down transformer X1, bridge rectifier BR1, 1µF capacitor C1 and 1,000µF capacitor C2, 12V voltage regulator 7812, 2.2kΩ resistor R1, and power indicator LED1. Transformer X1 reduces the input 230V AC to 15V AC, which is converted to DC by BR1 and filtered by the capacitors to get a regulated 12V DC supply through IC 7812.

Water-level sensor SEN18. It has ten exposed conducting tracks of which five alternate one act as power tracks and the other five as sensing tracks. When the sensor is submerged vertically in water, all the tracks get bridged due to water's conductivity and the sensor works as a mere resistor. The value of this resistor starts decreasing with decreasing water level. The sensor can be powered with a 3.3V or 5V DC power, depending on the available supply voltage.

Relay. It provides complete electrical isolation between the controlling circuit and the output circuit. Its contacts can handle a high current flow with total electrical isolation from delicate low-power electronic circuit. It is controlled/triggered by a low DC voltage. A 12V SPDT is used here to

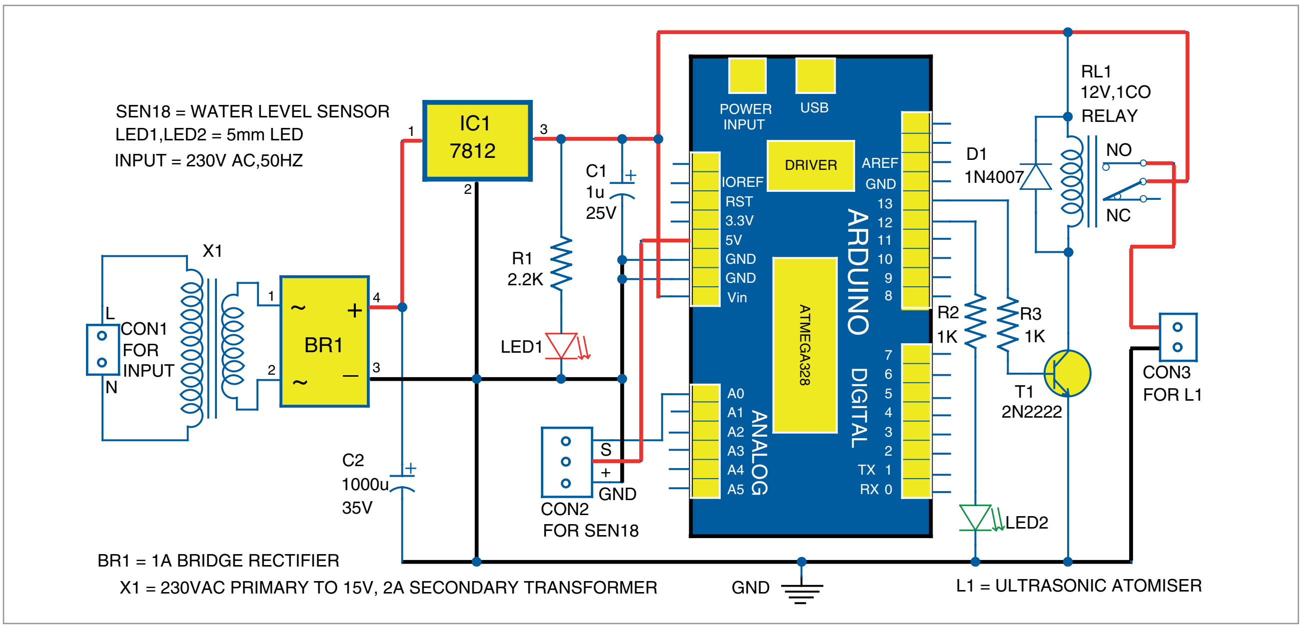

Fig. 1: Circuit diagram of condensed water atomiser